

Revision Notes for Class 7 Science Chapter 4 – The World of Metals and Non-metals

Metals and non-metals play a significant role in our daily life and have many interesting properties which make them valuable for different purposes. In this chapter, students will learn how to distinguish metals and non-metals based on daily life examples, experiments, and by exploring their physical and chemical properties. The world of metals and non-metals goes far beyond appearance—it influences everything from the crafting of tools in villages to making modern gadgets.

Properties of Metals and Non-metals

Metals usually have a shiny, lustrous appearance and are hard, such as iron, copper, and aluminium. However, not all metals are equally hard—sodium and potassium can even be cut with a knife. Mercury is exceptional as it remains in liquid form at room temperature. On the other hand, non-metals like sulfur, coal, and wood are dull and generally softer than metals. Students can observe these properties through simple activities, such as beating different materials with a hammer to test their malleability.

Malleability and Brittleness

Malleability is the property by which a material can be hammered into thin sheets without breaking. Metals like copper, aluminium, and iron are malleable and can be shaped into foils or sheets, which is why aluminium foil is used for food wrapping and thin silver foils decorate sweets. Gold and silver are the most malleable metals. Non-metals like coal and sulfur are brittle, breaking into pieces when struck, while wood is neither malleable nor brittle.

Ductility

Ductility allows metals to be drawn into wires. Copper and aluminium wires are commonly used in electrical circuits and stringed musical instruments. Gold is highly ductile; one gram can be stretched into a wire about two kilometers long! Non-metals such as coal and sulfur cannot be converted into wires as they lack ductility.

Sonority

When metals like a spoon or coin are dropped, they produce a ringing sound known as sonority. This is why bells, musical instruments, and even ghungroos use metals. Non-metals like coal and wood do not produce such sounds; instead, their sound is dull.

Conduction of Heat and Electricity

One of the key reasons why most cooking vessels are made of metals is their ability to conduct heat efficiently. In a simple experiment, a metal spoon placed in hot water gets hot quickly, whereas a wooden spoon does not. Metals are good conductors of heat and electricity, as demonstrated by the glowing of a bulb when materials like aluminium foil, iron nail, or copper wire are included in an electric circuit. Non-metals like wood, coal, sulfur, and rubber are poor conductors and do not let electricity pass through.

Effect of Air and Water: Rust and Corrosion

Iron objects develop brown deposits called rust when exposed to both air and water over time. This process, known as rusting, is a major problem as it weakens metal structures. Experiments show that iron does not rust when kept only in dry air or water; both air and water are necessary for rust to form. Corrosion is a broader term describing the gradual degradation of metal surfaces by air and water, such as green coatings on copper and black coatings on silver. Rusting can be slowed or prevented by painting, oiling, greasing, or galvanising (coating with zinc).

Reaction with Oxygen and Water: Oxides

Most metals form basic oxides on reaction with oxygen; for example, burning magnesium creates magnesium oxide, which turns red litmus paper blue. Some metals like sodium are so reactive they must be stored in kerosene to prevent reaction with moisture and oxygen. Non-metals, however, like sulfur, form acidic oxides when burned, which make blue litmus paper turn red. Non-metals generally do not react with water and, unlike metals, do not corrode easily.

Essential Uses in Everyday Life

Metals are not only used for utensils, cables, and tools, but their alloys are vital for atomic energy, transport, and modern technology. Their recycling is important for sustainability. Non-metals, despite their dull appearance, are essential for life—oxygen is required for breathing, nitrogen for plant fertilisers, chlorine for water purification, and iodine as an antiseptic.

Summary Table: Properties of Metals vs Non-metals

Property	Metals	Non-metals
Lustre	Usually lustrous	Usually non-lustrous
Malleability	Malleable	Not malleable, brittle
Ductility	Ductile	Not ductile
Conduction of Heat	Good conductor	Poor conductor
Conduction of Electricity	Good conductor	Poor conductor
Sonority	Sonorous	Not sonorous
Reaction with Oxygen	Forms basic oxides	Forms acidic oxides
Reaction with Water	Some react (e.g., sodium), most do not	Do not react
Corrosion	May corrode (e.g., iron rusts in moist air)	Usually resistant

Key Points for Revision

- Metals are mostly shiny, hard, malleable, ductile, sonorous, and good conductors of heat and electricity.
- Non-metals are generally dull, brittle, not malleable or ductile, and poor conductors.

- Rusting of iron happens only when both air and water are present; it can be prevented by coating, oiling, or galvanising.
- Metals form basic oxides; non-metals form acidic oxides with oxygen.
- Both metals and non-metals have important uses, from making tools to supporting life processes.

Application in Daily Life and Society

From the ancient Iron Pillar of Delhi, which has not rusted in centuries, to modern smartphones using multiple metals and non-metals, this chapter highlights the value, diversity, and applications of these elements. Understanding their properties helps explain their selection for specific tasks, encourages awareness about conservation, and sparks curiosity for further scientific discovery.